Computer Networks Project Proposal

IoT Telemetry Protocol (Sensor Reporting)

Presented to:

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1. Assigned Scenario

This project simulates a distributed IoT sensing environment, where multiple edge devices (clients) periodically send environmental measurements such as temperature and humidity to a central data collector (server) using the User Datagram Protocol (UDP).
Each device generates readings every second and transmits them along with metadata fields including:

- Device ID
- Sequence Number
- Timestamp
- Message type (INIT or DATA or other types will be specified in later versions)
- Other fields like Checksum and flags which will be specified in later phases and versions of our protocol.

The server acts as a collector that:

- Receives and logs all incoming packets.
- Maintains per-device state (last sequence number, timestamps).
- Performs duplicate suppression and loss detection.
- Detects out-of-order packets and sequence gaps.
- Records all data to a structured CSV log file for analysis.

2. Motivation

IoT systems often rely on lightweight, connectionless communication such as UDP due to its low overhead and simplicity; however, UDP does not guarantee reliable delivery, ordering, or duplication protection. To ensure data integrity, it is essential to implement reliability mechanisms at the application layer. We can study how reliability metrics behave under network stress and evaluate potential optimizations.

This project provides hands-on experience with:

- Network programming using sockets in Python.
- Understanding and mitigating packet loss, duplication, and delay.
- Designing and testing a custom reliability mechanism over UDP.
- Logging and analyzing real-time data streams from multiple concurrent clients.

3. Proposed Protocol Approach

Proposed system is composed of 2 main components: Client Devices (IoT Sensors) & Server

Client Behavior

- Can run multiple clients concurrently to simulate a network of devices. Each client represents a unique IoT sensor with a device_id.
- Sends temperature and humidity data every second for a configurable duration.
- Uses a per-device sequence number, incrementing by 1 with each message.
- Includes a timestamp for each transmission.
- The data is accompanied by Header (Total 11 bytes):

Header Field	Size (bits)	Description
Version	4	protocol version (version 1 for now)
MsgType	4	INIT or DATA (for our version 1 now)
DeviceID	16	Unique ID for each sensor
SeqNum	16	Order of sent packets (0 for INIT usually)
Timestamp	32	Current Unix timestamps (seconds)
Flags	8	Not used now in 1st version
Checksum	8	Not used now in 1st version

Server Behavior

- Listen to incoming UDP packets on a known IP and port.
- For each device, maintains the last seen sequence number.
- Perform duplicate suppression (same sequence number).
- Detect Lost packets (non-sequential jumps) & Out-of-order packets (timestamp earlier than expected).
- Log all received data to a CSV file